First Mid-Semester Exam **PHY301** (20 marks)

1. Consider the circuit equation for current I in a LCR circuit:

$$L\ddot{I} + R\dot{I} + I/C = 0$$

Given $L, C > 0$, what is the nature of the fixed point(s) as R is varied $(R \geq 0)$.

(5 marks)

2. Consider a conservative Newtonian system where a particle of mass $m = 1$ moves in a potential

$$V(x) = \frac{1}{2}x^2 - \frac{1}{4}x^4$$

(a) Enumerate the equilibrium (fixed) point(s) of the system.

(b) Find the nature of the fixed point through analysis of the Jacobian.

(c) Draw the phase portrait of the dynamics, and mark the sepratrix.

(d) What are the distinct types of motion that can take place in the system?

(8 marks)

3. Consider a system given by the equation: $\ddot{x} + 2\lambda\dot{x} + \omega^2 x = F$, where F is a constant.

Find the solution $x(t)$ for the system, for the case of $\lambda = \omega$, with initial conditions $x(0) = 0$, $\dot{x}(0) = 1$.

(7 marks)

Second Mid-Semester Exam PHY 301 (20 marks)

1. Consider a pendulum made of a spring with mass m on the end. The spring is arranged to lie on a straight line (which we arrange by, say, wrapping the spring around a rigid massless rod). The equilibrium length of the spring is l. Let the spring have length $l + x(t)$ and let the angle with the vertical be $\Theta(t)$. Assume the motion takes place in a vertical plane.

(a) Write down the Lagrangian.

(b) Write down the Euler-Lagrange equations.

(c) Find the equations of motion for x and Θ.

(6 marks)

2. Given a point mass m with Lagrangian of the form:

$$L(r, \theta, \dot{r}, \dot{\theta}) = \frac{m}{2}(\dot{r}^2 + r^2\dot{\theta}^2) + \frac{k}{r}$$

Find the Hamiltonian using Legendre Transforms.

(5 marks)

3. Show that

$$\frac{\partial^2 H}{\partial p^2} \cdot \frac{\partial^2 L}{\partial u^2} = 1$$

(3 marks)

4. A particle moves in a potential

$$V(r) = -V_0 e^{-(\lambda^2 r^2)}$$

(a) Write down the effective potential for the system, for a given angular momentum ℓ.

(b) Given ℓ, find the radius of the stable circular orbit. An implicit equation is fine.

(c) It turns out that if ℓ is too large, then no circular orbit exists. What is the largest value of ℓ for which a circular orbit does in fact exist?

(6 marks)